Activity 2

Observe a cockroach and identify its legs and wings. Try to know more about other parts of cockroach with the help of your teacher.

19.2.3 Birds

Birds can walk on the ground and fly as well. Some birds can also swim in the water. A bird has streamlined body. Its bones are light and strong. They are hollow and have air spaces between them. The lower portion of limbs are modified as claws, which help them to walk and to perch. The breast bones are modified to hold massive flight muscles which help in moving wings up and down. Birds have special flight muscles and the forelimbs are modified as wings.The wings and tail have long feathers, which help in flying. Birds show two types of flight: **gliding** and **flapping.**

During gliding the bird has its wings and tail spread out. In this movement the bird uses air currents for going up and down. Flapping is an active flight. The bird beats the air by flapping its wings. They use flight feathers for this purpose.



Figure 19.3 Movement in birds

Activity 3

Observe a hen and crow. How do they move? Write about the similarities and dissimilarities found among them, in your note book.

19.2.4 Snake

The body of snake consists of a large number of vertebrae. The adjoining vertebrae, ribs and skin are inter-connected with slender body muscles. When the snake moves, it makes many loops on its sides. The forward push of the loops against the surface makes the snake move forward. Movement of snake is called ***slithering movement***. Many snakes can swim in water also.



Figure 19.4 Movement in Snakes



Since snakes do not have legs, they use their muscles and their scales to move.

19.2.5 Fish

Fish swims with the help of fins. They have two paired fins and an unpaired fin. The body of a fish is streamlined to reduce friction while moving in water. They have strong muscles, which help in swimming. When a fish swims its front part curves to one side and the tail part stays in the opposite direction. In the next move, the front part curves to the opposite side and the tail part also changes its position to another side. The caudal or tail fin helps in changing direction.

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